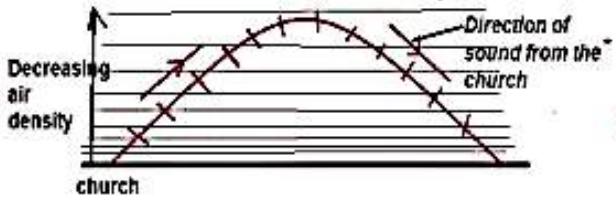
S.4 2026 PHYSICS

BUGANDA MOCKS

X/60

SCORE GUIDE

	Expected responses	Comment
ITEM ONE		
(a)	At night, *air near the ground is denser than that above.* ✓ Sound waves from the church travelling upwards undergoes *total internal reflection* ✓ and comes back to the ground making it seem louder  ✓ 03	
(b)	The dress appears *red.* ✓ In red light, *the red colour is reflected* ✓ because it is common and no colour is absorbed In yellow light, which *contains red and green* ✓, *Green colour is absorbed* ✓ and *red is reflected* ✓. 05	
(c)	$c = f\lambda$ Red: $3.0 \times 10^8 = f(7.5 \times 10^{-7})$ ✓ $\Leftrightarrow f = 4.0 \times 10^{14} m$ ✓ yellow: $3.0 \times 10^8 = f(5.8 \times 10^{-7}) \Leftrightarrow f = 5.2 \times 10^{14} m$ ✓ *red colour should be used since it has a small wavelength* ✓ 04	
	Total	12
ITEM TWO		

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	Total	12
ITEM TWO		
(a)	$(450 - 50) \longrightarrow 200 \longrightarrow 100 \longrightarrow 50 \longrightarrow 25 \longrightarrow 12.5 \longrightarrow 6.25$ ✓ $6t = T \Longrightarrow 6(2) = 12days$ ✓ The patient will remain in the hospital for 12 days ✓ 04	
(b)	Dangers of radioactive substances • Mutation ✓ • Eye irritation ✓ • Skin burn • Cancer • Infertility 02	Any two 1 score each
(c)	Handling radioactive substance • Use a pair of long tongs when carrying ✓ • Keep in a thick lead container ✓ • Wear protective gears • Keep away from if not in use • Don't eat/drink near it 02	Any two 1 score each
	Total	08

ITEM THREE		
	Moon appearance • The moon *reflects light from the sun*. ✓ • It *rotates/orbits the earth* and the amount of light it reflects depend on its phase. ✓ • These phases are; *new moon, waxing crescent, first quarter, waxing gibbous, full moon, wanning gibbous, third quarter and wanning crescent*. ✓ 03	

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ITEM THREE		
	Moon appearance • The moon *reflects light from the sun*. • It *rotates/orbits the earth* and the amount of light it reflects depend on its phase. • These phases are; *new moon, waxing crescent, first quarter, waxing gibbous, full moon, wanning gibbous, third quarter and wanning crescent*. 03	
	Live TV airing • It is possible due to *(communication) satellite* • Videos and sound were encoded(converted into electric signals) and sent to the ground station • They were then sent to the *(geosynchronous) satellite* through *uplink channel* and *amplified* • They were then sent back the ground station near the receiver through downlink channel • The signals were the sent to the TV decoder for *decoding* back into video and sound 05	
	Cause of day and night It is caused by *rotation of the earth* on its axis. The hemisphere of the earth *facing the sun experiences day time* The hemisphere of the earth *facing away from the sun experiences night time* 03	
	Why LED lights are preferred. • Affordable • Saves energy • Portable • Release less heat • Durable 02	Any two 1 score each

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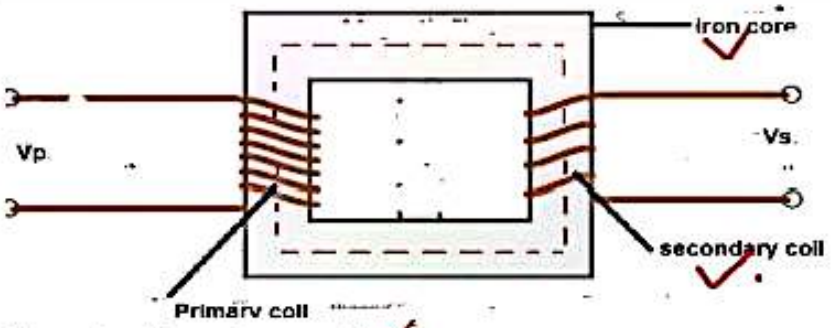
	Total	13
ITEM FOUR		
(a)	*Friction* force between the braking parts led to *heat generation* making the system feel hot. 02	
(b)	• The *bags loaded on the roof* *increased the C.O.G* making the car unstable on the road • Reduced by loading the bags in the *boot/inside the car* 03	
(c)	$v^2 = u^2 + 2as \Leftrightarrow 0^2 = 30^2 + 2a(30)$ $a = -15ms^{-2}$ $v = u + at \Leftrightarrow 0 = 30 + (-15)t$ $t = 2s$ The car tool 2 s to come to rest 04	
(d)	$Q = mc\Delta\theta = (30)(1200)(5) = 180{,}000J$	

	The energy absorbed by the system is 180,000J No heat is lost the surrounding 03	
	Total	12
ITEM FIVE		
(a)	$P = \frac{mgh}{t} \Rightarrow \frac{(1000)(0.200)(10)(8)}{60} = 266.7W$ The pump works at a rate of $266.7W$ 03	

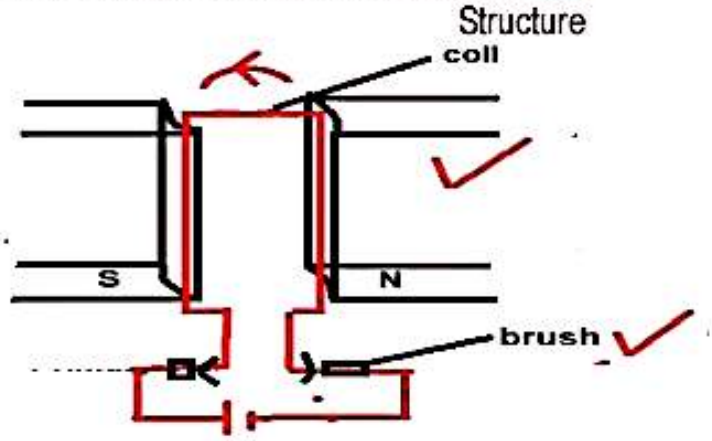
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	Total	12
ITEM FIVE		
(a)	$P = \frac{mgh}{t} \Rightarrow \frac{(1000)(0.200)(10)(8)}{60} = 266.7W$ The pump works at a rate of $266.7W$ 03	
	$Eff = \frac{work\ output}{work\ input} \times 100\%$ $Eff = \frac{266.7}{1800} \times 100\% = 15\% = 0.15$ The pump is not efficient since the ratio of useful work to energy input is less than 0.75 03	
	Due to *high SHC of water* which makes it to *heat up slowly during day time and slowly releases it at night.* By morning, *water still has some heat it absorbed the previous* day making it feel warm 03	
	Heat that evaporates water; $Q = ml_v = \left(\frac{600}{1000}\right)(2.56 \times 10^6) = 1{,}536{,}000J$ Heat that raise temp of evaporated water; $Q = mc\Delta\theta = \left(\frac{600}{1000}\right)(4200)(100 - 32) = 171{,}360J$ $1{,}536{,}000J$ evaporates water and $171{,}360J$ raises temp of evaporated water 03	
	Total	12

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	Total	12
ITEM SIX		
	The machine is a *step down transformer* ✓ $\frac{V_p}{V_s} = \frac{N_p}{N_s} \Leftrightarrow \frac{240}{150} = \frac{120}{N_s}$ ✓ $\Leftrightarrow N_s = 75$ ✓ The other coil has 75 turns ✓ 04	
	 *Ac enters the primary coil* ✓ This creates *changing magnetic field in the primary coil* ✓ The changing magnetic *field links up to the secondary coil.* ✓ A *lower voltage Vs is created in the secondary coil.* ✓ 06	
	$P = 960W = 0.960kW$ $Time = 4 \times 5 = 20hrs$ ✓ $Energy = 0.960 \times 20 = 19.2kWh$ ✓ $Cost = 19.2 \times 1000 = 19200/-$ ✓ The budget is not sufficient since its more than 10,000/- ✓ 04	
	Caring and maintaining a rechargeable battery. • Refilling of acid with water to maintain the acid level ✓ • Regular charging • Regular terminal cleaning • Avoid over charge/discharge • Avoid short circuit 01	Any 1 1 score each
	Total	15

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	• Avoid short circuit	
	Total	15
ITEM SEVEN		
	An electric drill contains an *electric motor*  Mode of operation • Current is fed into the coil *creating magnetic field around it* • These magnetic fields are *opposed* by those in the magnets • This causes *the coil to rotate* hence spinning the drill 06	
	$P = \frac{V^2}{R} \Rightarrow R = \frac{240^2}{12} = 96\Omega$ The drill has a resistance of 96Ω 03	
	factor / Solution Friction between moving parts generating heat and causing wear / lubrication Resistance in the coils leading to heat generation / Use thick, shorter copper wires Core loss/eddy currents leading to heat formation / Use iron core out of thin insulated iron layers Air resistance leading to drag / Use aerodynamic roters	Any two plus solution

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